Computer Science & IT

Data Structure & Programming



Queue

Lecture No. 01



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Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

postfix expression evaluation

prefix (Right to Left) top : operand$_1$
top-1 : operand$_2$

practice question

Slide

Topics to be Covered



Topic

Topic

Topic

Topic

Topic

Queue

Basic operation on queue.

Slide

Queue



Last
Rear

front (delete)

Queue is

↑ New entry
Insert

↑ CPU

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Topic : Queue

* Queue is 2 ended data structure
* front = delete, rear - Insert
* Logical property of queue First in first out.
* Application : CPU scheduling, Ready queue, Device Resource sharing (FIFO) order

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Topic : Queue

* front Index : Delete operation will happen Index pointed by front pointer
* Rear Index : Insertion will control by Index pointed by rear.
* Insertion in queue called **Enqueue** operation
* Deletion from queue called **dequeue** operation

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Topic : Queue

* for Enqueue
  * check full condition.
  * Increment the rear and Insert the value
  * if first Insertion then Increment front too.
* for dequeue
  * check Empty condition
  * Delete element from front Index Increment the front
  * Befor Increment check if Last elemen is deleted if yes assign both front & rear to -1

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Topic : Queue

Global variable

int a[100];
int front;
int rear;

IsEmpty return -1 if Queue is empty otherwise 0.

Zero

```
void init () {
    front = -1;
    rear = -1;
}
```

```
int IsEmpty () {
    if (front == -1)
        return 1;
    else
        return 0;
}
```

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Topic : Queue

```
void Enqueue( int x) {
    if (rear == 99) {
        printf("Queue is full");
        return;
    }
    rear = rear + 1;
    a[rear] = x;
    if (front == -1)
        front=0;
}
```

← check full condition

Increment rear & Insert value

→ if first Insertion
Increment front too.

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Topic : Queue

```
int Dequeue( ) {
    int data;
    if (front == -1){// if(IsEmpty())
        printf("Queue is Empty");
        return -1;
    }
    data = a[front];
    if (front == rear)
        front = rear = -1;
    else front = front+1;
    return data;
}
```

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Queue



Insert(1)



front = -1
rear = -1

front = 0
rear = 0



→ only for first Insertion

Insert(2)



front = 0
rear = 1



front = 0
rear = 0

Slide

Queue

Insert(3)



front = 0, rear = 1

front = 0
rear = 2

Insert(4)



front = 0
rear = 3



front = 0
rear = 2



Slide

Queue

Insert(5)



front = 0, rear = 3

Insert(6)



front = 0
rear = 4

front = 0
rear = 4



front = 0

rear = 4



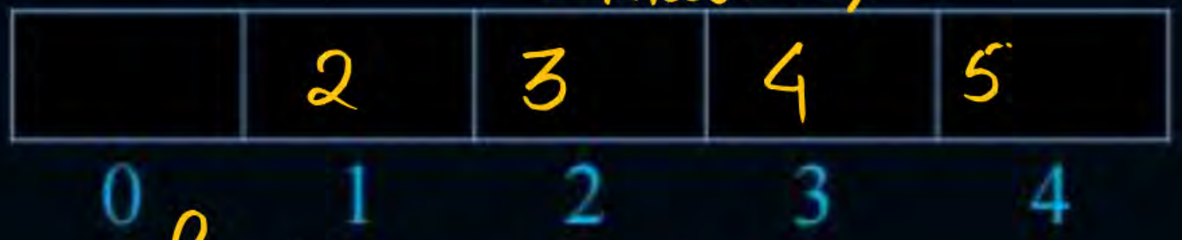
Queue is full.

Slide

Queue

Linear Queue

Insert(6)



front = 1, rear = 4



front = 1, rear = 4 overflow





Slide

Queue

Delete



front = 0
rear = 4

Delete



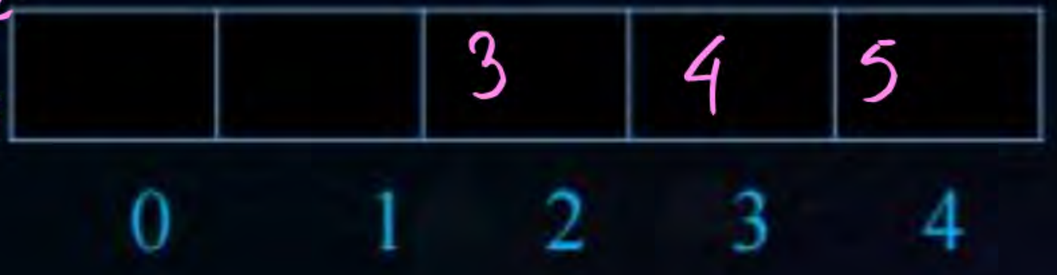
front = 1
rear = 4

front = 1



rear = 4

front = 2



rear = 4

Slide

Queue

Delete()



front = 2 rear = 4

Delete()



front = 3

rear = 4



front = 4

rear = 4



Slide

Queue



front = 4
rear = 4



front = -1
rear = -1



Last element deleted
front = rear



Slide



Queue









Slide



Topic : Queue

Suppose Size of queue is 10
& following operation are performed



Enqueue(10), Enqueue(20), Enqueue(25), X = dequeue()
Enqueue(95), Y = dequeue(), Z = dequeue(), Enqueue(26)

20 → Y
25 → Z
X = 10

X + Y - Z will be

10 + 20 − 25 = 5

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Topic: GATE 2021 Set-I, 1-Mark

Consider the following sequence of operations on an empty stack.

```
push (54); push (52); pop (); push (55); push (62); s = pop ();
```

Consider the following sequence of operations on an empty queue.

```
enqueue (21), enqueue (24), deque (); enqueue (28); enqueue (32); q = dequeue();
```

The value of s+q is _____

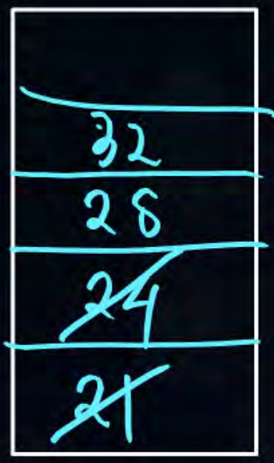
(A) 90
(B) 82
(C) 86
(D) None

Stack



S = 62





q = 24

62 + 24 = 86

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Topic: GATE 2023, 2-Marks

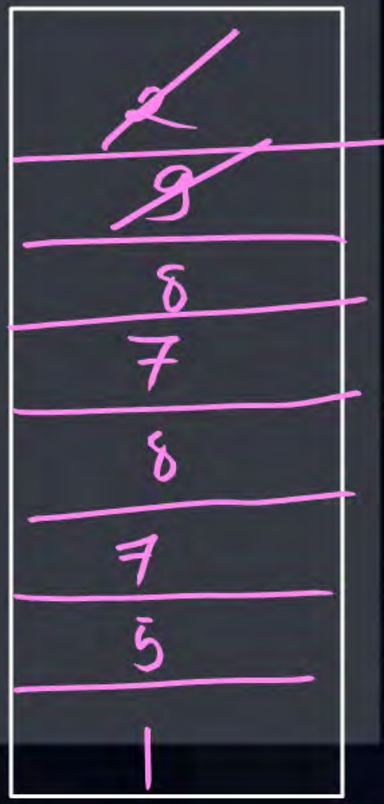
Consider a sequence a of elements $a_0 = 1, a_1 = 5, a_2 = 7, a_3 = 8, a_4 = 9$, and $a_5 = 2$. The following operations are performed on a stack S and a queue Q, both of which are initially empty.

I. push the elements of a from a_0 to a_5 in that order into S.
II. enqueue the elements of a from a_0 to a_5 in that order into Q.
III. pop an element from S.
IV. dequeue an element from Q.
V. pop an element from S.
VI. dequeue an element from Q.
VII. dequeue an element from Q and push the same element into S.
VIII. Repeat operation VII three times.
IX. pop an element from S.
X. pop an element from S.

The top element of S after executing the above operations is _____.

8





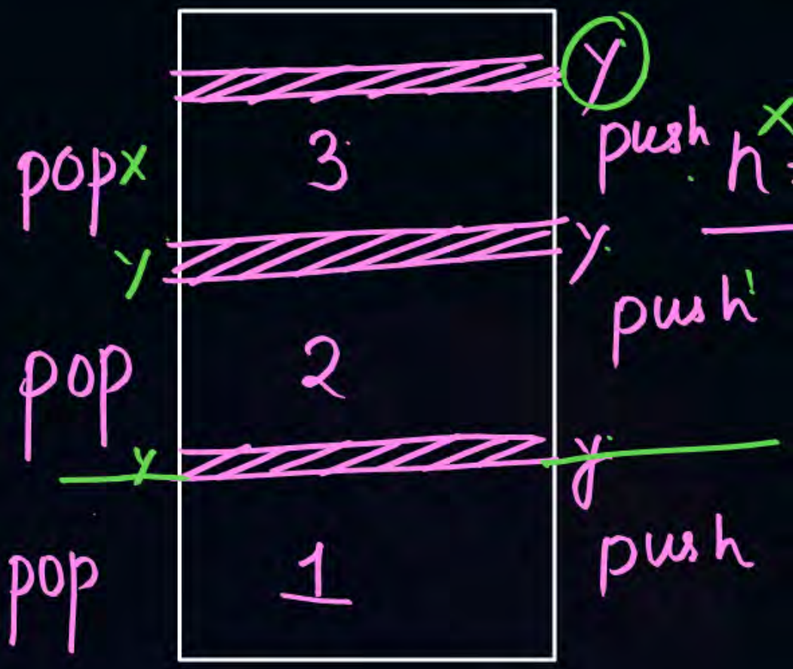
S



Q

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Topic: Queue



1 elem (y)

push pop = x

y - time elapsed between end of one stack & start of another.

3 — (y) (n = 1)

Stack Life for 2 : 3y + 2x

Stack Life for 1 : 4x + 5y

Avg = $\frac{y + 3y + 2x + 4x + 5y}{3} = \frac{9y + 6x}{3} = \boxed{3y + 2x}$

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2 mins Summary

Topic — Queue

Topic — Enqueue

Topic — Dequeue

Topic

Topic

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THANK - YOU

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